

CROSS-USER CORRELATION AND PAIRING-CONTEXT METHODS

A. Overview

In certain embodiments, the computing system supports privacy-preserving correlation between a first user's metrics and a second user's metrics. Cross-user correlation may be performed without requiring explicit anatomical imagery and without explicit anatomical labeling. Cross-user correlation may be supported across a range of pairing contexts including male-to-female, male-to-male, female-to-female, and other pairings.

B. Cross-User Enrollment and Consent Gating

In various embodiments, cross-user correlation is enabled only after user authorization. Authorization may include explicit consent from each user, account linking, time-limited permissions, revocable access controls, and/or consent scopes that limit which metrics may be used for correlation. In some embodiments, users may choose one-way sharing, two-way sharing, and/or anonymized sharing.

C. Privacy-Preserving Representation for Correlation

Cross-user correlation may use privacy-preserving representations derived from raw signals and/or features, including indices, ranges, zones, curves, time-window summaries, and confidence indicators. In some embodiments, raw signals are not shared between users; instead, the system computes correlation using derived metrics and/or compressed representations. In certain embodiments, the computing system uses normalization, quantization, hashing, or other transformations to reduce identifiability while retaining correlation utility.

D. Alignment, Synchrony, and Stability Metrics

In various embodiments, cross-user correlation includes computing one or more of: 1. an alignment score indicating similarity or compatibility of abstracted metrics; 2. a timing synchrony score indicating alignment of temporal signatures or response timing across users; 3. a stability score indicating repeatability, confidence, and/or robustness of the correlation across time windows; 4. a context-matched score computed within matched states such as sleep, rest, low-motion segments, or other validated windows; and/or 5. a confidence indicator computed from signal quality indices, wear validation, artifact rejection, and/or measurement recency. Correlation may be computed using similarity measures, correlation coefficients, lag analysis, distance metrics, learned embeddings, and/or other approaches.

E. Pairing Context Inclusion

In certain embodiments, the system supports cross-user correlation across different pairing contexts without changing the privacy-preserving output format. Pairing contexts may include male-to-female, male-to-male, female-to-female, and other pairings. In some embodiments, the correlation computation is context-agnostic and operates on abstracted metrics; in other embodiments, the system selects different metric subsets or weighting functions based on user-selected pairing context while maintaining privacy-preserving outputs.

F. Cross-Node Inputs for Cross-User Correlation

In various embodiments, cross-user correlation may incorporate inputs from multiple nodes, including: external longitudinal physiology metrics, intraluminal longitudinal physiology metrics, external geometry metrics, intraluminal geometry and compliance metrics, and upper-body context metrics. In certain embodiments, cross-user correlation requires minimum data quality thresholds and may prefer context-matched windows. In certain embodiments, baseline acquisition results are used to normalize correlation inputs and to improve repeatability across sessions.

G. Output and User Interface Examples

The system may output cross-user correlation results as privacy-preserving representations including alignment scores, synchrony curves, stability indicators, confidence indicators, and/or zone-based summaries. In certain embodiments, the system provides explanatory tags indicating whether correlation is computed using matched context windows, whether confidence is reduced due to artifact rejection, and/or whether additional data collection is recommended to improve stability.

H. Variations and Alternatives

Cross-user correlation may be performed locally, on a mobile device, or in a cloud environment. In some embodiments, correlation is performed on-device with encrypted exchange of derived metrics; in other embodiments, correlation is performed on a server with access controls. In some embodiments, cross-user correlation supports revocation, expiration, audit logs, and/or user-configurable privacy modes.